Swati Aggrawal

swatiagg357@gmail.com | github.com/swatiaggrawal | LinkedIn

EDUCATION

Sapienza Università di Roma **Expected: May 2026**

Master's in Computer Science

- **Coursework:** Deep Learning And Applied AI, NLP, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning (Ongoing)
- Selected for a competitive international master's program

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity To Computers In Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort)

RESEARCH EXPERIENCE

Machine Learning for Air Pollution — May 2023

- Published a peer-reviewed **IEEE ViTECoN-2023** paper evaluating RF, SVM, GA, LS-SVM, and ANN for air quality prediction, comparing model suitability, identifying research gaps, and improving performance over traditional methods.

Continual Learning for Computer Vision — May 2025

- Reviewed **60+ papers** on catastrophic forgetting, replay- and regularization-based methods, and TIL/DIL/CIL protocols in continual learning, synthesizing best practices, open challenges, and a structured reference framework for future CV research.

PROJECTS

Lossy Text Compressor

- Implemented **ML pipeline** for semantic text compression using **Transformers**
- Evaluated compression–quality trade-offs using **BLEU, ROUGE and cosine similarity**
- Built modular, reproducible experiments in **Python**

DocDelta

- Built and deployed a **production-ready ML inference API using FastAPI** for scalable semantic document comparison
- **Dockerized** NLP pipeline and deployed on **AWS ECS (Fargate)**
- Designed **stateless REST endpoints** for semantic document comparison
- Optimized CPU-only PyTorch inference for **low-latency cold starts**
- Implemented **cloud monitoring** using AWS CloudWatch

SKILLS

- **Programming:** Python, Bash, Linux
- **ML Engineering:** Machine Learning, Model Evaluation, Feature Engineering, CNNs, Transformers
- **ML Deployment:** FastAPI, REST APIs, Docker, AWS ECS, Fargate, CloudWatch
- **Frameworks:** PyTorch, TensorFlow, Keras
- **Tools:** Git, GitHub